

UNIVERSITY OF LISBON

ISEG- LISBON SCHOOL OF ECONOMICS AND MANAGEMENT

NORMAL SEASON EXAM - JANUARY 2023

ADVANCED ECONOMETRICS

Module Convenor:
Paulo M.D.C. Parente

Use answer booklets of ISEG (folhas de teste do ISEG)

Instructions (please read before starting): Write in a clear legible manner in ink/ballpoint. Do not use pencils or erasable pens. Approved calculators are permitted. Only one sheet (2 sides A4) of notes made exclusively by the student may be consulted (no material distributed by the teacher in any form is allowed). Whenever conducting a test use a 5% significance level unless stated otherwise. Also be sure to state null and alternative hypotheses, null distribution (with degrees of freedom), rejection criterion (critical values and rejection region) and outcome. If you are asked to derive something, give all intermediate steps also. Do not answer questions with a “yes” or “no” only, but carefully justify your answer.

Section A - Topics in Microeconometrics

Question 1

Let t_i^* denote the duration of some event, such as unemployment, measured in continuous time. Consider the following model for t_i^* :

$$\begin{aligned} t_i^* &= \exp(\mathbf{X}_i' \boldsymbol{\beta}_0 + u_i), \quad u_i | \mathbf{X}_i \sim N(0, \sigma^2), \\ t_i &= \min\{t_i^*, c\} \\ &= \begin{cases} t_i^* & , \text{ if } t_i^* < c \\ c & , \text{ if } t_i^* \geq c \end{cases} \end{aligned}$$

where $c > 0$ is a known censoring constant and $\mathbf{X}_i$ is a vector of regressors. Let $\phi(\cdot)$ and $\Phi(\cdot)$ be the density function and the cumulative distribution function respectively of the standard normal random variable.

- (a) (1.5 Mark) Find the mathematical expression for $P(t_i = c | \mathbf{X}_i)$ as a function of $\Phi(\cdot)$.
- (b) (1.5 Mark) Find the mathematical expression for density function of $\log(t_i)$ (given $\mathbf{X}_i$) when $t_i < c$ as a function of $\phi(\cdot)$. [**Hint-** You may use the following result: If W is a normal random variable with mean μ_W and variance σ_W^2 , its density function is given by $f_W(w) = \sigma_W^{-1} \phi\left(\frac{w - \mu_W}{\sigma_W}\right)$]
- (c) (1.5 Mark) Write down the log-likelihood function for observation i .
- (d) (2.5 Marks) If $z \sim N(0, 1)$ it can be shown that

$$E(z | z < a) = -\frac{\phi(a)}{\Phi(a)}$$

for any a . Use this fact to prove that

$$E(\log(t_i) | \log(t_i) < \log(c), \mathbf{X}_i) = \mathbf{X}_i' \boldsymbol{\beta}_0 - \sigma \frac{\phi\left(\frac{\log(c) - \mathbf{X}_i' \boldsymbol{\beta}_0}{\sigma}\right)}{\Phi\left(\frac{\log(c) - \mathbf{X}_i' \boldsymbol{\beta}_0}{\sigma}\right)}.$$

- (e) (3 Marks) Use the result in (d) to prove that

$$E(\log(t_i) | \mathbf{X}_i) = \Phi\left(\frac{\log(c) - \mathbf{X}_i' \boldsymbol{\beta}_0}{\sigma}\right) [\mathbf{X}_i' \boldsymbol{\beta}_0 - \log(c)] + \log(c) - \sigma \phi\left(\frac{\log(c) - \mathbf{X}_i' \boldsymbol{\beta}_0}{\sigma}\right).$$

- (f) (1 Mark) Consider the following method for estimating $\boldsymbol{\beta}_0$. Using all observations, we run the Ordinary Least Squares regression of $\log(t_i)$ on $\mathbf{X}_i$. Explain why this does not generally produce a consistent estimator of $\boldsymbol{\beta}_0$.
- (g) (1 Mark) Obtain the log-likelihood function if the censoring time is potentially different for each person, so that $t_i = \min\{t_i^*, c_i\}$, where c_i is observed for all i . Assume that u_i is independent of $(\mathbf{X}_i, c_i)$.

Section B - Topics in Time Series

Question 2

Let

$$X_t = \phi X_{t-1} + \varepsilon_t,$$

and

$$Y_t = \eta_t + \theta \eta_{t-1},$$

where ε_t and η_t are independent white noise processes with variances σ_ε^2 and σ_η^2 respectively, and $|\phi| < 1$. Let $Z_t = X_t + Y_t$.

(a) (2 Marks) Show that

$$Z_t = \phi Z_{t-1} + v_t,$$

where $v_t = \varepsilon_t + \eta_t + (\theta - \phi) \eta_{t-1} - \phi \theta \eta_{t-2}$.

(b) (2 Marks) Obtain the autocorrelation function of v_t . Given the obtained result, what is the ARMA model of Z_t ?

Question 3

(2 Marks) Let

$$\begin{aligned} Q_t &= \alpha Q_{t-1} + D_t, \\ D_t &= \beta D_{t-1} + \varepsilon_t \end{aligned}$$

where $\{\varepsilon_t\}$ is a white noise process with variance σ_ε^2 and $|\beta| < 1$. What is the ARMA model of Q_t ?

Question 4

(2 Marks) Consider the stochastic process $W_t = \beta t + \varepsilon_t$ where $\{\varepsilon_t\}$ is a white noise process with variance σ_ε^2 . Prove that W_t is not an $I(1)$ process.

[END OF PAPER]